

Measuring What a Human Adds to a Machine Baseline: A Problem Statement

Aaron Sun, September 2026 (revised 17 September). Working note, not a paper.

1. The question

Three studies asked the same question in different domains: when a human departs from a machine baseline, does the departure carry anything the baseline lacks? In chess (Sun, 2026a) the baseline is a strictly stronger engine; human deviations have a direction, and the direction is not repaid. In forecasting (Sun, 2026b) the baseline is a fallible model; the human median carries information the best single model does not encompass, and who has that information is a stable individual attribute. In noisy image classification (Sun, 2026c, a locked-specification reanalysis of ImageNet-16H and Tejeda et al., 2022) the baseline is a fine-tuned classifier; individual human labels carry information the best classifier lacks, the increment survives an ensemble of the other classifiers at every one of 20 classifier variants, and people who see the classifier's prediction improve but do not beat it.

The three studies used four operationalizations. Whether they measure the same thing has not been stated. This note states it, so that the next study can be designed against one quantity rather than four.

2. One quantity

Let o be an outcome, m the baseline's output, h the human's output, and L a proper loss. Let F be a family of functions that combine m and h , and F_0 the subset of F that ignores h . Define the human's increment over the baseline as

$$I(h \mid m) = \min_{g \in F_0} E[L(g(m), o)] - \min_{f \in F} E[L(f(m, h), o)]$$

the reduction in expected loss from adding h to the best use of m alone. The first term is the recalibrated baseline, not the raw one. If the raw baseline were used, the gain from calibrating m would be counted as information from h . This is the value of h as additional information given m . It is non-negative in population because F contains F_0 , so its sign is never the finding. The finding is its size, what it is relative to, and where it lives.

Three refinements are needed before $I(h \mid m)$ is a usable target.

Relative to what. Any input that is not pure noise has positive I . The forecasting study's reference-curve diagnostic made this concrete. When the human median was replaced by the median of the other 33 models, the reference was above the public median on the encompassing coefficient at every point of the quality range, and above the superforecaster median at the top of the range; elsewhere the two were not distinguishable. So the quantity of interest is the distinctive increment

$$D(h \mid m) = I(h \mid m) - I(r \mid m)$$

where r is a reference source of the same kind the deployer could have used instead of a human (another model, an ensemble). Only superforecasters had $D > 0$ on the scale-free measure. A study that reports I without a reference reports how much any second opinion would help, not how much a human would.

D answers a substitution question: if one more source can be added, should it be a human or another model. The oversight question is different: with the best machine source r already in use, does h still add anything. That is $I(h \mid m, r)$, the increment of h over the best use of m and r together. D can be negative while $I(h \mid m, r)$ is positive, because h and r can be complementary. The next study should report both. One caveat carries over from the paper. The reference was the median of 33 forecasts and the human median was the median of about 7 or 49, so a human curve below the reference mixes distinct information with extra noise. A reference matched in size, the median of as many models as there were forecasters on the target, would remove that.

Whose. $I(h \mid m)$ can be computed per person. The forecasting study found per-forecaster increment stable across a split of targets (split-half $r = 0.78$ for superforecasters, 0.64 for the public); the image study found $r = 0.49$ across a split of images. If the individual increment is a trait, the deployment question changes from "should a human oversee this" to "which humans".

Under what conditions. $I(h \mid m)$ should depend on the baseline's error. Where the baseline is right, there is nothing to add. The forecasting study varied the baseline across 34 models and found the increment did not vanish as the baseline improved, within the observed quality range; but that range was narrow and the result is not yet a curve. The image study varied it across 20 classifier variants spanning a sixfold range of log loss, and the increment given the ensemble never fell to zero.

3. How the four operationalizations map onto it

Encompassing coefficient (forecasting, H2). The coefficient on $\text{logit}(h) - \text{logit}(m)$ in a logistic regression of o on $\text{logit}(m)$ and that difference. This is a test of $I > 0$ within the logistic combining family, under log loss. It is a sign test, not an estimate of size, and its scale moves with the variance of $h - m$, which is why it could not be compared across baselines of different quality.

Cross-fitted log-score gain (forecasting, v2 robustness). The out-of-sample reduction in log loss from adding h to a logistic combiner that already contains $\text{logit}(m)$. This is a direct estimate of $I(h \mid m)$ in the logistic family, with the recalibrated baseline as the first term, on the natural scale. It is the quantity the encompassing coefficient stands in for, and it should be the primary measure going forward.

RES (chess, §7.2 redesign). The opponent's realized error after the human's move, minus the error predicted from the difficulty of the resulting position. This is not $I(h \mid m)$ in the sense above, because in chess the human's "output" is an action, not a belief, and the relevant loss is realized over an opponent's reply. RES measures something narrower: whether the human's deviation exploits a component of value that the engine's objective does not represent (the opponent's fallibility). Formally, if V_m is the engine's value function (against a perfect opponent) and V^* the value against the actual opponent population, RES tests whether the human's choice loads on $V^* - V_m$.

Cross-fitted log-loss gain with a confusion-row human vector (image classification). The human gives a label and a confidence level, not a probability vector. The vector used is the cross-fitted row of a confusion matrix, $P(\text{true class} \mid \text{label, confidence})$, smoothed toward the same row pooled across noise levels (a change from the specification's original uniform

Dirichlet prior, recorded as a dated amendment after the pilot and before the full run), the human-side model of Kerrigan, Smyth and Steyvers (2021). $I(h | m)$ is then the out-of-sample reduction in 16-class log loss from adding that vector to a two-weight log-linear pool with the recalibrated classifier as the no-human term. This is the same quantity as the forecasting measure, computed directly. On 900 test images and 145 people, $I(h | a)$ is +0.13 nats per item for an individual and +0.22 for the pooled humans on an item; against the ensemble of the other four architectures at the same fine-tuning level, $I(h | a, r_a)$ is +0.11 and +0.20, both with intervals excluding zero, and neither falls to zero across the 20 variants.

$I(h | m)$ itself can be computed in chess if o is set to the deep-engine value of the position after the move and h is treated as a candidate move the combiner may choose. That was not done in Sun (2026a). It would put the chess increment on the same scale as the other two.

That gives the distinction the three domains actually differ on. A baseline can be wrong in two ways: estimation error (a wrong belief about the outcome) and objective error (a right belief about the wrong objective). Forecasting models have estimation error; the human increment there is about the outcome. A depth-15 engine has estimation error relative to deeper search, but it is small next to the error of a titled player in a fast game. What the engine carries that the player does not is an objective error, the perfect-opponent assumption; the human increment there, if any, would be about the objective. The chess study found the human's deviation is directional (it does load on sharpness) but not profitable (the loading does not translate into $V^* - V_m$). The forecasting study found the human's belief is informative about the outcome. The image study found the same, and more strongly: there the human's information is not what an ensemble of models supplies. The three domains line up as follows.

Table 1. Three domains.

	Chess	Forecasting	Image classification
Baseline	engine, strictly stronger	best of 34 language models	best of 20 classifiers
Human beats baseline on average	no	superforecasters yes, public no	no (individual), yes (pooled)
Human carries information beyond baseline	no (RES slope not positive)	yes, both groups (encompassing)	yes, individual and pooled
Survives an ensemble reference	not tested	superforecasters only, scale-free measure	yes, individual and pooled
Increment located where baseline is least certain	not tested	no	not tested
Per-person increment stable	not tested	$r = 0.78 / 0.64$	$r = 0.49$
Seeing the model	not tested	not tested	raises the gain at every level; does not make people beat the

The chess and image cells are not the same test: chess measures RES, the others measure $I(h | m)$ directly. Nor are the ensemble references the same strength. In forecasting the reference is the median of 33 models from different vendors; in image classification it is four architectures at the same fine-tuning level, trained on the same data and the same noisy images, and therefore far more correlated with the baseline. A weaker reference leaves more for the human to add, so part of the run from nothing (chess) through experts only (forecasting) to everyone (images) is a difference in references, not only in domains. The pattern that remains is that the human adds something where the baseline makes errors the human can see.

Two statements follow, one provable and one not. The bound: $I(h | m)$ cannot exceed the reducible loss of the baseline, the gap between the recalibrated baseline's loss and the loss of the best predictor available from all information. The conjecture: the share of that gap a human recovers depends on which kind of error the baseline makes. Estimation error was recoverable by good forecasters. Objective error in chess was not recovered by titled players in fast time controls. Whether it is recoverable by anyone, or whether the engine's objective error is too small at depth 15 to matter, the chess data cannot say.

4. What the measure does not capture

The chess and forecasting studies measured $I(h | m)$ with humans who never saw m ; the image study's main results did too. Its Tejada part is the one measurement of the seen case in this line, and it has no no-model control arm, so it cannot separate the interaction term from the order in which trials were seen. That is the independent-information case. In deployment the human sees the baseline, and h changes when it does. The change can go either way. Automation bias and algorithm aversion move h toward or away from m without adding information; attention drawn to the points of disagreement can add information that an independent forecast would not contain. So the independent case is a benchmark for what oversight can add, not a bound on it. The difference between the two is a separate quantity, call it the interaction term, and neither study in this line measured it.

Two results from outside this line bear on it. Steyvers, Tejada, Kerrigan and Smyth (2022) model the complementarity of independent human and model predictions and estimate the combination from data, which is the independent case above with a fitted combiner. Vaccaro, Almaatouq and Malone (2024) pool experiments in which people worked with a model in view and find that the combination is on average worse than the better of the two alone, with gains concentrated where the human alone outperformed the model. Read against the quantity above, that pattern says the interaction term is often negative and may depend on $I(h | m)$. It is the quantity most relevant to whether human oversight of AI systems is effective, and it needs an experiment where the same people forecast with and without seeing the model, with a control arm that sees nothing.

5. Open questions, in the order they should be attacked

1. Is $I(h | m, r) > 0$ for anyone other than superforecasters? In forecasting, no: the public's increment was not distinguishable from an ensemble's. In image

classification, yes, for ordinary crowd workers. So "human oversight adds value" is not reducible to "expert oversight adds value"; which people add value depends on the domain, and the next question is what distinguishes the two domains. One candidate answer is perceptual access: the image task gives the human a channel the classifier does not have at that noise level, the forecasting task does not. The competing answer is reference strength: the image ensemble is four correlated architectures, the forecasting ensemble is 33 models from different vendors. Separating the two needs an image ensemble built from independently trained models, or a forecasting ensemble restricted to one vendor.

2. Is the individual increment stable across time, not only across a split of one round? This needs a second wave of the same forecasters.

3. What does the increment curve look like across a wider range of baseline quality? The 2024 model generation spans a Brier ratio of 2.5x; the question needs a range where the baseline goes from clearly worse than the human to clearly better.

4. Can objective error be separated from estimation error in a single domain, so that the two kinds of increment are measured on the same people? Chess with a deliberately weakened engine (shallow depth, so that estimation error appears) and a deep engine as the outcome would do this, and would measure the same quantity as the forecasting study.

5. What is the interaction term, and does it depend on the individual increment? A person with high $I(h | m)$ who defers to the model when shown it has an oversight value of zero. The image study gives a first reading from Tejeda et al. (2022), sequential design (each person answers, sees the classifier, and may revise the same image; 72 participants after exclusions). Seeing the classifier raised people's gain over it at every classifier level, but at no level did people who saw the classifier beat it. The slope of what survives on the person's prior increment, estimated as $\beta_{\text{seen}} - \beta_{\text{unseen}}$ on split halves of the unassisted trials, was negative at all three levels without an interval excluding zero. The concurrent design was blocked rather than interleaved and is reported as descriptive only. The direction matches the deference account; the power does not settle it.

Question 1 now has one positive and one negative case; question 2 waits on a second wave; question 5 has a first, underpowered reading. Question 5 is the one that matters for policy and the one that needs a designed experiment.

References

Kerrigan, G., Smyth, P., & Steyvers, M. (2021). Combining human predictions with model probabilities via confusion matrices and calibration. *Advances in Neural Information Processing Systems*, 34.

Steyvers, M., Tejeda, H., Kerrigan, G., & Smyth, P. (2022). Bayesian modeling of human-AI complementarity. *Proceedings of the National Academy of Sciences*, 119(11), e2111547119.

Sun, A. (2026a). Sharper, not safer: The direction of human deviations from engine play. doi.org/10.5281/zenodo.22267000

Sun, A. (2026b). Where the benchmark can err: Human deviations from model forecasts and their return. doi.org/10.5281/zenodo.22589676

Sun, A. (2026c). Human increment over image classifiers, against an ensemble reference and with the model shown: a locked-specification reanalysis of ImageNet-16H and Tejeda et al. (2022). github.com/aaronsun923/hai-reference

Tejeda, H., Kumar, A., Smyth, P., & Steyvers, M. (2022). AI-assisted decision-making: A cognitive modeling approach to infer latent reliance strategies. *Computational Brain & Behavior*, 5, 491–508.

Vaccaro, M., Almaatouq, A., & Malone, T. (2024). When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8, 2293-2303.